COOKIE CATS

A/B Test — Product Retention Analysis

Gate 30 vs Gate 40 | Day-7 retention

19.02%	18.20%	+0.82 pp
Gate 30 Day-7 retention	Gate 40 Day-7 retention	Observed difference

Business Question

Does moving the first game gate from level 30 to level 40 affect player retention?

Cookie Cats tested two versions of the game: Gate 30, where the first progression gate occurs at level 30, and Gate 40, where it occurs at level 40. The primary metric for this analysis is Day-7 retention.

Executive Summary

The Gate 30 group recorded higher Day-7 retention than the Gate 40 group. The observed difference was 0.82 percentage points. A two-proportion test produced a p-value of 0.0016, and the analytical 95% confidence interval was +0.31 to +1.33 percentage points.

The result is a retention finding, not a complete product or financial evaluation. The available data does not include monetization or long-term engagement measures.

1. Experiment Setup

The analysis compares two experiment groups and uses Day-7 retention as the primary KPI.

Item	Description
Control	Gate 30
Treatment	Gate 40
Primary KPI	Day-7 retention
Secondary KPI	Day-1 retention
Engagement metric	Gamerounds
Initial dataset	90,189 players
Final analysis set	90,188 players

Experiment group sizes

Experiment	Players	Share
Gate 30	44,699	49.56%
Gate 40	45,489	50.44%

2. Data Preparation

Before analysing the experiment, the dataset was checked for missing values, duplicate users, duplicate records, data types, experiment group composition and extreme gameround values.

The dataset had no missing values and no duplicate user IDs. One unusually large gameround value of 49,854 was identified. The median was 16 gamerounds and the second-highest value was 2,961.

The extreme observation was excluded from the main analysis and then tested through a sensitivity check. The conclusion remained essentially unchanged.

Analysis	Difference	p-value
Excluding extreme value	+0.818 pp	0.001592
Including extreme value	+0.820 pp	0.001554

3. Initial Findings

Day-7 retention was higher in the Gate 30 group.

Metric	Gate 30	Gate 40
Day-1 retention	44.82%	44.23%
Day-7 retention	19.02%	18.20%
Day-7 retained players	8,501	8,279

Observed Day-7 difference: +0.82 percentage points.

At this stage, the difference is descriptive. The next step is to determine whether the observed difference is consistent with more than random variation.

4. Statistical Test

Day-7 retention is a binary outcome: a player either returned or did not return. A two-proportion test was therefore used to compare the retention proportions between the two groups.

Measure	Result
Null hypothesis	Day-7 retention is the same for Gate 30 and Gate 40.
Alternative hypothesis	Day-7 retention differs between Gate 30 and Gate 40.
Significance level	5%
Z-statistic	3.16
p-value	0.0016

5. Confidence Interval

The estimated difference in Day-7 retention was +0.82 percentage points. The 95% confidence interval was +0.31 to +1.33 percentage points.

The interval provides a range around the estimated difference and shows the uncertainty associated with the estimate. It is useful alongside the p-value because it describes the size of the difference, not only the evidence for a difference.

6. Bootstrap Validation

A bootstrap procedure with 5,000 simulations was used as an additional check on the uncertainty estimate.

Measure	Result
Bootstrap simulations	5,000
Bootstrap mean difference	+0.823 pp
Bootstrap 95% CI	+0.32 to +1.34 pp
Analytical 95% CI	+0.31 to +1.33 pp

The bootstrap interval was very close to the analytical confidence interval, providing a useful robustness check on the estimated uncertainty.

7. Business Impact

Measure	Result
Observed retained-player difference	222
Scaled estimate per 100K players	≈818
Relative difference vs Gate 40	4.50%

The 818 figure is a scaled estimate, not an observed experiment result.

8. Engagement Analysis

Players were grouped by gamerounds into Low (8 or fewer), Medium (9–35), and High (more than 35) engagement segments.

Segment	Gate 30	Gate 40	Difference
Low	1.54%	1.64%	-0.09 pp
Medium	8.68%	7.97%	+0.71 pp
High	47.66%	46.38%	+1.27 pp

The segment results show different observed retention patterns, but they should not be interpreted as proof that the gate change has different causal effects for different player types.

Gamerounds were measured during the experiment. This makes the segmentation exploratory rather than a clean pre-experiment characteristic. The interaction analysis also did not provide sufficient evidence that treatment effects differed significantly by segment.

Interaction	p-value
Medium engagement	0.122
High engagement	0.225

9. Product Context and Limitations

The analysis evaluates retention, but the dataset does not contain the measures needed to evaluate the broader product impact.

Available in the dataset	Not available
Day-1 retention	Revenue
Day-7 retention	Purchases / monetization
Gamerounds	Long-term retention
Experiment group	Long-term progression outcomes

The analysis establishes a difference in Day-7 retention between the two gate configurations, but it does not establish the overall financial impact of moving the gate.

A broader product decision would need to consider retention alongside monetization, progression and longer-term player behaviour.

10. Main Finding

Gate 30 had higher Day-7 retention than Gate 40 by 0.82 percentage points. The difference was statistically significant, with a p-value of 0.0016 and a 95% confidence interval of +0.31 to +1.33 percentage points.

The result remained essentially unchanged when the extreme gameround observation was included.

11. Conclusion

The A/B test provides evidence of a difference in Day-7 retention between the Gate 30 and Gate 40 configurations. Gate 30 recorded 19.02% Day-7 retention compared with 18.20% for Gate 40.

The observed difference was +0.82 percentage points, with a 95% confidence interval of +0.31 to +1.33 percentage points and a p-value of 0.0016. The result was robust to the extreme gameround observation identified during data validation.

The available dataset does not include monetization or long-term engagement measures. As a result, the retention finding should be considered one input into a broader product assessment rather than a complete measure of the impact of the gate change.

Project Deliverables

Deliverable	Purpose
Excel workbook	Data cleaning, calculations, analysis and dashboard
Python analysis	Statistical testing, bootstrap, segmentation and sensitivity analysis
Dashboard	Stakeholder-facing summary of the experiment
Case study	Written explanation of the analysis and findings

Tools: Excel • Python • pandas • NumPy • SciPy • statsmodels • matplotlib

Prepared from the completed Cookie Cats analysis and dashboard work.